

# Nitin Gupta

Kathmandu, Nepal

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GitHub: github.com/Neettin — LinkedIn: linkedin.com/in/nitin-gupta

## Professional Summary

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Machine Learning Engineer with hands-on experience building, evaluating, and deploying end-to-end ML systems. Strong in data preprocessing, imbalanced learning, model optimization, and production-ready AI applications using Python and modern ML frameworks.

## Technical Skills

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**Languages:** Python, SQL, HTML, CSS

**ML & AI:** Scikit-learn, TensorFlow, Deep Learning, NLP, Time Series Analysis

**Data:** Pandas, NumPy, Matplotlib, Seaborn

**Tools & Platforms:** Flask, FastAPI, Streamlit, Power BI, Git, GitHub, Jupyter, Google Colab

**Concepts:** ML Pipelines, Feature Engineering, Model Evaluation, Imbalanced Data Handling, Deployment, Data Structures & Algorithms

## Projects

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**Credit Card Fraud Detection System** [GitHub](#) — [Live Demo](#) — [Notebook](#)

- **Problem:** Detecting fraudulent transactions in highly imbalanced financial data with 0.17% fraud cases.
- **Solution:** Applied SMOTE, NearMiss, and ensemble methods with precision-recall optimization.
- **Result:** Achieved **95% precision** and **93% recall** using Random Forest with cross-validation.

**Solanix AI – Potato Disease Detection** [GitHub](#) — [Live Demo](#)

- **Problem:** Manual inspection of potato crops for Early and Late Blight is time-consuming and error-prone.
- **Solution:** Built a CNN-based image classification system with data augmentation and transfer learning.
- **Result:** Achieved **98.61% validation accuracy** and **97.66% test accuracy** on leaf disease classification.

**Movie Recommendation System** [Live Demo](#)

- **Problem:** Users struggle to discover relevant movies from large catalogs.
- **Solution:** Developed content-based filtering using TF-IDF vectorization and cosine similarity.
- **Result:** Delivered real-time personalized recommendations with **92% relevance accuracy** based on user testing.

**Sentiment Analysis Platform (SentimentX)** [Live Demo](#)

- **Problem:** Businesses need automated tools to analyze customer feedback at scale.
- **Solution:** Built an NLP pipeline using TF-IDF and Logistic Regression for multi-class sentiment classification.

- **Result:** Achieved **89% accuracy** in classifying text into positive, negative, and neutral categories.

#### **Diamond Price Prediction** GitHub

- **Problem:** Diamond pricing is complex with multiple interacting features affecting value.
- **Solution:** Created regression pipeline with feature encoding, scaling, and ensemble methods.
- **Result:** Achieved **R2 score of 0.97** using XGBoost, outperforming traditional regression models.

#### **Education**

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##### **BSc (Hons) Computer Systems Engineering (BSc.IT)**

International School of Management and Technology, Kathmandu

Oct 2022 – Nov 2025

##### **Higher Secondary Level (+2)**

Modern Public Secondary School, Nepalgunj

May 2020 – Sep 2022

#### **References**

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Sandeep Pokhrel, Senior Lecturer — Phone: +977 9851357792